Week 16 Self-Assessment

Regression, R-Squared, and Prediction

Gordon Wright

1 Instructions

Test your understanding of regression analysis, the meaning of R-squared, and how we use regression for prediction. Regression builds on correlation to allow us to predict one variable from another.

---

2 Part 1: Conceptual Questions

2.1 Question 1: What is Regression?

Simple linear regression is used to:

- (A) Compare two groups
- (B) Predict one variable from another using a linear equation
- (C) Test for differences before and after
- (D) Determine if variables are categorical

In regression, the predictor variable is also called:

- (A) The dependent variable
- (B) The independent variable (IV) or X variable
- (C) The outcome
- (D) The residual

The outcome variable is also called:

- (A) The predictor
- (B) The dependent variable (DV) or Y variable
- (C) The independent variable
- (D) The slope

2.2 Question 2: The Regression Equation

The regression equation is: Y = a + bX. In this equation:

'a' (the intercept) represents:

- (A) The slope of the line
- (B) The predicted Y value when X = 0

- (C) The correlation coefficient
- (D) The sample size

'b' (the slope) represents:

- (A) The Y-intercept
- (B) The change in Y for each one-unit increase in X
- (C) The total variance
- (D) The residual

2.3 Question 3: Regression vs Correlation

How is regression different from correlation?

- (A) They are exactly the same
- (B) Regression allows prediction and has a specific direction (X predicts Y)
- (C) Correlation is more accurate
- (D) Regression only works with experiments

If the correlation between X and Y is strong, the regression will:

- (A) Fail to work
- (B) Make more accurate predictions
- (C) Always predict perfectly
- (D) Have a slope of zero

2.4 Question 4: R-Squared in Regression

R-squared (R2) tells us:

- (A) The significance level
- (B) The proportion of variance in Y explained by X
- (C) The slope of the line
- (D) The sample size needed

If R2 = .36, this means:

- (A) The correlation is .36
- (B) 36% of participants showed the effect
- (C) 36% of the variance in Y is explained by X
- (D) The prediction is 36% accurate

The remaining variance (1 - R2) is called:

- (A) Explained variance
- (B) Unexplained or residual variance
- (C) Total variance
- (D) The intercept

2.5 Question 5: Residuals

A residual is:

- (A) The predicted value
- (B) The slope
- (C) The difference between the actual Y value and the predicted Y value
- (D) The correlation coefficient

If a residual is positive, the actual score was:

- (A) Lower than predicted
- (B) Higher than predicted
- (C) Exactly as predicted
- (D) Cannot determine

2.6 Question 6: Assumptions of Regression

Linear regression assumes:

- (A) A linear relationship between X and Y
- (B) A curvilinear relationship
- (C) No relationship at all
- (D) Categorical variables only

If the relationship is clearly curved (nonlinear), we should:

- (A) Use linear regression anyway
- (B) Consider transforming variables or using nonlinear methods
- (C) Ignore the pattern
- (D) Collect more data

---

3 Part 2: Interpretation Questions

3.1 Question 7: Interpreting Regression Output

A regression analysis yields: Y = 25 + 3X, R2 = .42

If X = 10, the predicted Y is:

- (A) 25
- (B) 30
- (C) 55
- (D) 42

For every 1-unit increase in X, Y increases by:

- (A) 25
- (B) 3
- (C) 42
- (D) 10

3.2 Question 8: Interpreting R-Squared

Study A reports R2 = .64. Study B reports R2 = .16.

Which study's predictor explains more variance?

- (A) Study A
- (B) Study B
- (C) They are equal
- (D) Cannot compare

In Study A, what percentage of variance is NOT explained?

- (A) 64%
- (B) 36%
- (C) 100%
- (D) 16%

3.3 Question 9: From Correlation to R-Squared

If r = .60, what is R2?

- (A) .60
- (B) .36
- (C) 6.0
- (D) .30

If r = -.50, what is R2?

- (A) -.50
- (B) -.25
- (C) .25
- (D) .50

Note: R-squared is always positive because we square the correlation.

3.4 Question 10: Interpreting the Slope

A regression predicting exam score from hours studied shows: Score = 50 + 5(Hours)

This means:

- (A) Everyone gets 50
- (B) Each additional hour of study is associated with 5 more points
- (C) Studying 5 hours gives a score of 50
- (D) The maximum score is 55

A student who studies 8 hours is predicted to score:

- (A) 58
- (B) 50
- (C) 90

- (D) 80

---

4 Part 3: Application Questions

4.1 Question 11: Making Predictions

A researcher develops a regression equation to predict stress (Y) from workload (X): Stress = 10 + 2.5(Workload), R2 = .49

For someone with a workload score of 8, predicted stress is:

- (A) 10
- (B) 20
- (C) 30
- (D) 49

How much variance in stress is explained by workload?

- (A) 10%
- (B) 25%
- (C) 49%
- (D) 2.5%

4.2 Question 12: Evaluating Prediction Quality

A regression with R2 = .81 will make:

- (A) Poor predictions
- (B) Good predictions (81% variance explained)
- (C) Perfect predictions
- (D) No predictions

A regression with R2 = .09 will make:

- (A) Excellent predictions
- (B) Poor predictions (only 9% variance explained)
- (C) Moderate predictions
- (D) Perfect predictions

4.3 Question 13: Limitations of Prediction

We should be cautious about:

- (A) Making any predictions
- (B) Predicting beyond the range of observed data (extrapolation)
- (C) Using regression at all
- (D) Large R-squared values

If our data has X values ranging from 1-10, predicting Y when X = 50 is:

- (A) Very reliable

- (B) Unreliable extrapolation
- (C) More accurate than interpolation
- (D) Impossible

---

5 Part 4: MCQ Exam Practice

5.1 Investigation 1

A researcher examines whether hours of practice predicts piano performance scores. Data from 40 students yields: Performance = 30 + 4(Practice Hours), R2 = .52, $p < .001$.

1. **What type of analysis is this?**

- (A) t-test
- (B) Correlation only
- (C) Simple linear regression
- (D) ANOVA

2. **Predictor (IV):**

- (A) Practice hours
- (B) Performance score
- (C) Sample size
- (D) R-squared

3. **Outcome (DV):**

- (A) Practice hours
- (B) Performance score
- (C) The slope
- (D) The intercept

4. **For each additional hour of practice, performance increases by:**

- (A) 30 points
- (B) 4 points
- (C) 52 points
- (D) 1 point

5. **A student who practices 10 hours is predicted to score:**

- (A) 30
- (B) 40
- (C) 70
- (D) 100

6. **How much variance in performance is explained?**

- (A) 30%
- (B) 4%

- (C) 52%
- (D) 70%

5.2 Investigation 2

A study finds that sleep quality (X) predicts daily mood (Y) with R2 = .25.

1. **What percentage of variance in mood is explained by sleep quality?**

- (A) 75%
- (B) 25%
- (C) 50%
- (D) 5%

2. **What is the correlation (r) between sleep and mood?**

- (A) .25
- (B) .50 (since R2 = .25, r = sqrt(.25) = .50)
- (C) .75
- (D) .125

3. **What percentage of variance is NOT explained?**

- (A) 25%
- (B) 75%
- (C) 50%
- (D) 100%

4. **This unexplained variance could be due to:**

- (A) Other factors affecting mood (exercise, social contact, stress, etc.)
- (B) Errors in R-squared
- (C) The sample being too large
- (D) Sleep quality being irrelevant

5. **Can we say sleep quality CAUSES better mood?** TRUE / FALSE

5.3 Investigation 3

A regression equation is: Test Score = 40 + 0.5(Study Minutes), R2 = .36

1. **The intercept (40) represents:**

- (A) Predicted score when study time = 0
- (B) The slope
- (C) R-squared
- (D) The maximum score

2. **Studying for 60 minutes predicts a score of:**

- (A) 40
- (B) 60
- (C) 70

- (D) 100

3. **Studying for 120 minutes predicts a score of:**

- (A) 60
- (B) 80
- (C) 100
- (D) 120

4. **A student actually scored 80 after studying 60 minutes. Their residual is:**

- (A) −10
- (B) +10 (actual 80 minus predicted 70)
- (C) 0
- (D) 80

5. **The positive residual means they scored:**

- (A) Higher than predicted
- (B) Lower than predicted
- (C) Exactly as predicted
- (D) We cannot tell

5.4 Investigation 4

A researcher wants to predict job satisfaction from salary. R2 = .15.

1. **How would you evaluate this R2?**

- (A) Excellent - salary fully explains satisfaction
- (B) Modest - salary explains only 15% of variance in satisfaction
- (C) R2 is too small to interpret
- (D) Perfect prediction

2. **What does this suggest about other factors?**

- (A) No other factors matter
- (B) 85% of variance is due to other factors (work environment, colleagues, autonomy, etc.)
- (C) Salary is the only factor
- (D) We cannot speculate

3. **Should we conclude salary doesn't matter for satisfaction?**

- (A) No - it still explains some variance; other factors also matter
- (B) Yes - 15% is too small
- (C) The study is invalid
- (D) We need a larger sample

4. **Multiple regression could help by:**

- (A) Removing the effect of salary

- (B) Including additional predictors to explain more variance
- (C) Making R2 exactly 1.0
- (D) Eliminating residuals

---

6 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain the purpose of regression analysis
- ☐ Interpret the regression equation (Y = a + bX)
- ☐ Understand what the intercept (a) and slope (b) represent
- ☐ Calculate predicted values using a regression equation
- ☐ Interpret R-squared as proportion of variance explained
- ☐ Convert between r and R-squared
- ☐ Explain what residuals represent
- ☐ Recognise limitations of prediction (extrapolation)

! Key Distinction

Correlation tells us about the strength and direction of a relationship. Regression goes further by providing a specific equation for prediction and quantifying how much variance is explained. However, regression (like correlation) still cannot establish causation without an experimental design.